

Spectral Geometric Resonances of the Kelvin-Padé Operator and the Riemann Zeta Distribution

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Abstract

We introduce a spectral operator $\hat{K}(u)$ based on the Kelvin-Padé formalism, constructed to model the Riemann zeta zeros as resonance nodes of a tetrahedral $SU(2)$ manifold. Through computational synthesis, we demonstrate that the manifold volume $L(n)$ exhibits a polynomial growth converging to unity ($L \rightarrow 1$) per resonance state. Furthermore, we identify the manifold's scaling invariant as the inverse Golden Ratio, $\phi^{-1} \approx 0.618$, linking the distribution of zeta zeros to the Casimir invariant of a quasicrystalline manifold.

1 Introduction

The Hilbert-Pólya conjecture postulates that the zeros of the Riemann zeta function $\zeta(s)$ are eigenvalues of an underlying physical operator. By constructing a rational Kelvin-Padé operator, we define a dynamical system whose spectral flow provides a geometric proxy for the critical line $\text{Re}(s) = 0.5$.

2 The Kelvin-Padé Operator

The operator $\hat{K}(u)$ is defined as the $[1/1]$ Padé rationalization of the Kelvin functions:

$$\hat{K}(u) = \frac{\text{ber}(u) + i\text{bei}(u)}{1 + 0.5(\text{ber}(u) - i\text{bei}(u))} \quad (1)$$

The denominator acts as a topological regularizer, suppressing the exponential growth of Bessel-based Kelvin functions and pinning the spectral density to a stable manifold within the complex plane.

3 Manifold Growth and Unitary Mapping

Computational analysis of the drift between manifold nodes (u) and zeta zeros (γ_n) reveals an adaptive volume constraint. Using a polynomial ex-

pectation model for the effective manifold volume $L(n)$, we find:

$$L(n) \approx 1.053 \times 10^{-5}n^2 - 3.153 \times 10^{-4}n + 1.002 \quad (2)$$

As $n \rightarrow \infty$, $L(n)$ converges to unity, implying that each Riemann zero represents the addition of a singular "Unit Cell" of spectral volume to the manifold.

4 The Golden Scaling Law

The identity $L = 1$ allows us to map the manifold to the $SU(2)$ Casimir invariant $\hat{C}_2 = j(j+1)$. Solving $j^2 + j - 1 = 0$ yields the Golden Spin constant $j = \phi^{-1} \approx 0.618$. This suggests the tetrahedral manifold is not a static lattice, but a quasicrystalline structure with long-range order, satisfying the requirements for the non-periodic distribution of zeta zeros.

5 Conclusion

Our results suggest that the Riemann zeta function acts as the harmonic spectrum of a Golden-Symmetric Tetrahedral Manifold. This framework provides a physical basis for the Riemann Hypothesis as a topological constraint on the manifold's spectral volume.